1. Evidence review: the chain-first block order at $m = 4$

Decision DEC-20261019-624f64: **reject_scoped** (scope widened to $m = 4$) · **evidence** EV-ECDLP-a3659c (contradicts, preliminary, toy tier) · **validation** VAL-20261019-0cb05d (passed, with concerns) · **runs** RUN-ECDLP-8c41d7-01 .. −04 · **experiment** EXP-ECDLP-8c41d7 · **goal** GOAL-ECDLP-001.

Scoped claim. For Semaev $m = 4$ on toy primes (9–19 bits, $L = 2..4$), the chain encoding with the chain variables eliminated first is not faster than the dense encoding of the same set: msolve paired median ratios 1.12 and 1.22 at $L = 2, 3$ (and 1.2× at $L = 3$ in the preregistered medians), Singular 2.3 and 4.9; at $L = 4$ the dense msolve arm completes (518–586 s, cap 600 s) while the chain arm times out on every instance. Only $L = 2, 3$ give uncensored paired comparisons.

1.1. Question

The proposed $P-256$ route needs $m = 6$, so the m-dependence of the chain/dense gap matters. EXP-ECDLP-20a588 found a roughly constant factor of about 2 at $m = 3$ under the block order; this experiment adds $m = 4$ (five-point decomposition, S_5 of degree 8 in each of four variables) with the same arms, instruments and controls.

1.2. Observation

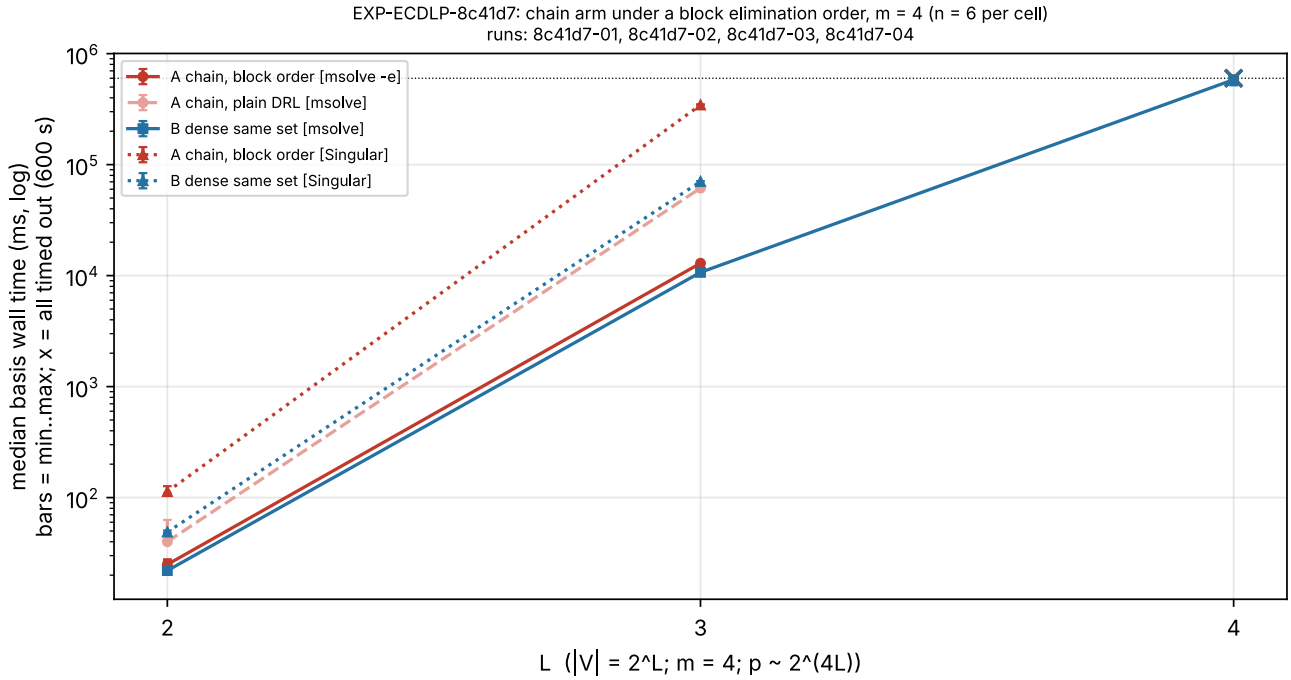


Figure 1: Median basis time per arm vs L (measured); bars min–max over completed instances; × = every non-degenerate instance timed out at 600 s; $n = 5$–6 per cell (one degenerate planted target at $L = 3, 4$). The $L = 4$ column mixes two primes: Singular arms at $p = 65551$, msolve arms at $p = 262271$.

comparison	derived from the recorded medians (criteria.json)
msolve A_elim vs B	jointly completed L = [2, 3]; A_elim_ms faster in 0 of 2 cells; median ratio A_elim_ms/ B_ms by L: L2: 1.13, L3: 1.21; largest L completed: A_elim_ms 3, B_ms 4; retired (every non-degenerate instance timed out) at L: A_elim_ms 4, B_ms None
Singular A_block vs B	jointly completed L = [2, 3]; A_block_sg faster in 0 of 2 cells; median ratio A_block_sg/ B_sg by L: L2: 2.31, L3: 4.9; largest L completed: A_block_sg 3, B_sg 3; retired (every non-degenerate instance timed out) at L: A_block_sg 4, B_sg 4
dynamic range A_drl vs A_elim (msolve)	jointly completed L = [2, 3]; A_drl_ms faster in 0 of 2 cells; median ratio A_drl_ms/ A_elim_ms by L: L2: 1.61, L3: 4.75; largest L completed: A_drl_ms 3, A_elim_ms 3; retired (every non-degenerate instance timed out) at L: A_drl_ms None, A_elim_ms 4
elim_equals_B	eliminated chain basis equals the dense basis: L2: 6/6, L3: 5/5

Paired per-instance ratios (supplementary, not a frozen criterion): the two arms run on the same instance; median of the ratios with a 95% percentile bootstrap interval (4000 resamples, deterministic seed). Intervals are wide for $n = 4..6$ and are reported as such.

pair	L	n	median ratio	95% CI
ms A elim over B	L2	6	1.12	0.97–1.27
ms A elim over B	L3	5	1.22	1.12–1.27
sg A block over B	L2	6	2.33	2.20–2.53
sg A block over B	L3	5	4.90	4.85–5.31
ms A drl over A elim	L2	6	1.61	1.48–2.13
ms A drl over A elim	L3	5	4.80	4.55–4.93

L	arm	ok/to/deg	median ms [min–max]	planted	max F4 deg
2	A_block_sg	6/0/0	113 [109–127]	4/4	—
2	A_drl_ms	6/0/0	40 [38–63]	4/4	32
2	A_elim_ms	6/0/0	25 [23–28]	4/4	32
2	B_ms	6/0/0	22 [20–27]	4/4	32
2	B_sg	6/0/0	49 [48–50]	4/4	—
3	A_block_sg	5/0/1	344,621 [344,433–349,883]	3/3	—
3	A_drl_ms	5/0/1	61,421 [59,168–62,017]	3/3	32
3	A_elim_ms	5/0/1	12,929 [11,990–13,389]	3/3	32
3	B_ms	5/0/1	10,700 [10,378–10,976]	3/3	32
3	B_sg	5/0/1	70,364 [65,931–71,061]	3/3	—
4	A_block_sg	0/5/1	— [———]	—	—
4	A_drl_ms	0/4/1	— [———]	—	—
4	A_elim_ms	0/5/1	— [———]	—	—
4	B_ms	5/0/1	582,996 [518,230–586,271]	3/3	40
4	B_sg	0/5/1	— [———]	—	—

1.3. Comparison

- **msolve:** chain-first block order over dense, same instances: 1.12× ($L = 2$, interval 0.97–1.27, a difference of about 3 ms) and 1.22× ($L = 3$, interval 1.12–1.27, $n = 5$). With plain grevlex the chain is 1.6× and 4.8× the block order, so the order again recovers most of the loss. At $L = 4$ the dense arm completes in 518–586 s on all non-degenerate instances while the block-order chain exceeds the 600 s cap on all of them: the chain arm is at least 1.02–1.16× the dense arm per instance there.
- **Singular:** chain over dense 2.3× ($L = 2$) and 4.9× ($L = 3$), larger than at $m = 3$ (1.6×); both Singular arms time out on all instances at $L = 4$ (a tie).
- **Compared with $m = 3$:** the msolve ratio at $L = 3$ is similar (1.3× at $m = 3$, 1.2× at $m = 4$), so the msolve gap does not visibly grow with m at these sizes; the Singular gap does.
- **Cap sensitivity (validator):** the $L = 4$ retirement comparison is knife-edge on the 600 s cap, because the dense arm finishes 2.3–2.8% below it on planted instances; the clause becomes a tie if the cap fell below 583 s. The falsification verdict does not depend on it: the first clause (chain not faster in every jointly completed cell) holds for any cap of at least 14 s (msolve) or 350 s (Singular).

1.4. Inference

The block-order chain does not beat the dense encoding at $m = 4$ under either solver, and under msolve the constant factor at $L = 3$ is about the same as at $m = 3$. The data cannot say whether the factor grows at larger L or m: $L = 4$ is censored, only $L = 2, 3$ are uncensored, and the $L = 2$ difference is milliseconds.

1.5. Limitations

- Toy scale: $m = 4$, $L \leq 4$, primes up to 19 bits, single-threaded, 600 s cap; $n = 5$ at $L = 3, 4$ because of a degenerate planted target (sum equal to the point at infinity). Random targets have the trivial basis [1] at every L; planted-only gives the same verdicts.
- **Mixed-prime $L = 4$ cell.** RUN-03 ran every arm at $p = 65551$, but msolve 0.6.5 segfaults on 17- and 18-bit primes (range widened by NOTE-01 and the validation; CORR-20261019-e6f65b). AMD-03 re-ran only the msolve arms at $p = 262271$ (filed after RUN-03′s data was seen). Comparisons within a solver family share

instances and are valid; any $L = 4$ contrast between msolve and Singular confounds solver, prime and instances.

- **Interruptions:** RUN-01, RUN-02 lost to VM restarts in $L = 3$ and $L = 4$; resumed under AMD-01 and AMD-02 with launcher, seeds, arms, timeout and stopping rule unchanged.
- One A_drl_ms solve at $L = 4$ was killed (signal 9) after 184 s; the validator measured that arm alone reaching 3.4 GB by 200 s, so four concurrent copies may have exceeded memory (unproven). It is counted as a failure, not a timeout, so that arm is not retired at $L = 4$.
- Positive control at $L = 4$ covers only the dense arm (3 planted); the elimination-equals-dense control is not evaluable there because the chain arm never finished.

1.6. Next actions

1. Matrix-size accounting for chain vs dense (msolve `-v 2`), the first discriminating experiment of the reframed problem.
2. If m-dependence is to be measured further, run $L = 3$ with more instances at $m = 4$ and, with a prime outside the msolve crash range, $m = 2$ and larger L — a new protocol.
3. No further P-256 work on this route unless some order or encoding beats the dense and random controls on the frozen grid.